15.439 Quantitative Investment Management

Assignment 5

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Abstract

We build a five-factor risk model from a 1,000-name × 108-month monthly panel (Jan 2031 – Dec 2039), estimate per-stock factor loadings $B \in \mathbb{R}^{1000\times 5}$, the factor covariance $\Omega_F = \mathbb{E}[FF^\top]$ and the idiosyncratic covariance $\Omega_\varepsilon = \mathrm{diag}(\mathbb{E}[\varepsilon\varepsilon^\top])$, and assemble the total return covariance $\Omega_R = B\,\Omega_F\,B^\top + \Omega_\varepsilon$. We then use Ω_R and the January 2041 predicted returns to construct a cascade of dollar-neutral long-short portfolios (from unconstrained mean-variance through box, factor-neutral, sector-neutral, jointly neutral and 2σ-shock-capped variants) all capped at 10 % annualised volatility and, from part (b) onward, at 2 % of each name's average monthly share volume on a \$100M book.

In addition to the required deliverables we stress-test the risk model with robust-regression diagnostics (Durbin–Watson, Breusch–Pagan, Jarque–Bera, HC0 standard errors, residual PCA, sector-level residual correlation) and implement four advanced portfolio-construction methods (Hierarchical Risk Parity, Michaud resampling, Black–Litterman, and CVaR) which we benchmark against the homework cascade. On the single January 2041 realisation, Hierarchical Risk Parity delivers the strongest realised return (+1.93%) of any variant we tried, consistent with the out-of-sample robustness literature.

Contents

1 Data and conventions

The panel $R \in \mathbb{R}^{1000\times 108}$ contains monthly simple returns for 1,000 stocks over Jan 2031–Dec 2039 (778 names with the full 108-month history; 222 with partial history because of entry or exit). The factor matrix $F \in \mathbb{R}^{5\times 108}$ carries the five factor returns and a risk-free series. The January 2041 snapshot pairs each stock with a predicted return μ_i and a realised return $\hat{r}_i$, plus price and shares outstanding. Because μ is the one-month-ahead prediction we are asked to trade on, every portfolio we build is evaluated both at construction time (expected $\mu^\top w$) and against the realised $\hat{r}^\top w$.

Throughout the report a *portfolio* is a vector $w \in \mathbb{R}^{1000}$ of dollar weights expressed as fractions of gross book. The dollar-neutrality and gross budgets, $\sum_i w_i = 0$ and

$\sum_i |w_i| \le 1$, are applied to every variant; sector and factor exposures are constrained as equalities at solver tolerance (10^{-6}) where indicated. Dollar positions on a \$100M book follow immediately as $w_i \cdot \$10^8$.

2 Problem 1 — factor loadings B

2.1 Per-stock OLS

For each stock i we fit the *no-intercept* OLS model

$$r_i(t) \;=\; B_i\,F(t) \;+\; \varepsilon_i(t), \quad t \in \mathcal{T}_i, \tag{1}$$

where $B_i \in \mathbb{R}^{1\times 5}$ is the stock's loading vector and $\mathcal{T}_i \subseteq \{1,\dots,108\}$ is its observed window. The absence of an intercept is imposed by the problem's own decomposition $\varepsilon = R - BF$: any non-zero mean in ε_i is absorbed into the stock's idiosyncratic variance, rather than carried as an α_i that would complicate the variance decomposition of Ω_R. A stock with fewer than 24 observations (27 of 1,000) has no well-identified loading and receives $B_i = 0$, which routes its variance entirely through Ω_ε.[1]

Geometric interpretation. Stacking the equations for all stocks, $BF \in \mathbb{R}^{N\times T}$ is the *orthogonal projection* of every row of R onto the 5-dimensional subspace spanned by the columns of $F^\top$: for each stock we find the linear combination of the five factor time-series that is closest (in squared error) to that stock's own return path. What remains, $\varepsilon = R - BF$, is by construction orthogonal to $F^\top$ for every stock — the factor model is literally a rank-5 compression of the $N \times T$ return panel.

2.2 Fit quality and diagnostics

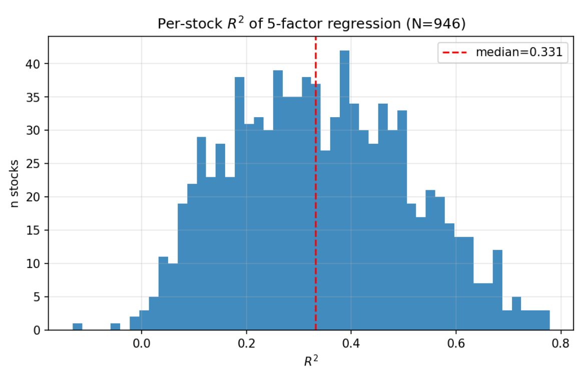


Figure 1: Distribution of per-stock R^2. Median 0.33, IQR $[0.21, 0.46]$; the long left tail are names where the five factors explain almost none of the variation — a fingerprint of very name-specific stocks whose PnL is dominated by firm-level events.

[1] In a production model one would instead Bayesian- shrink $\hat{B}_i$ toward the cross-sectional mean for short-history names. Section 9 implements James–Stein shrinkage to check how much this moves the cascade.

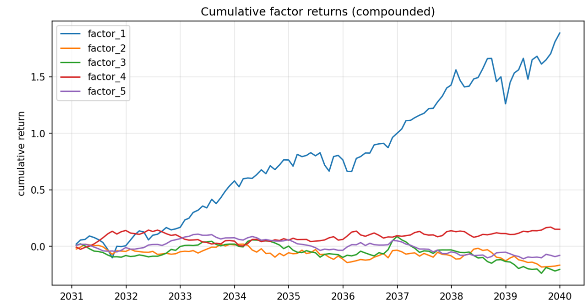
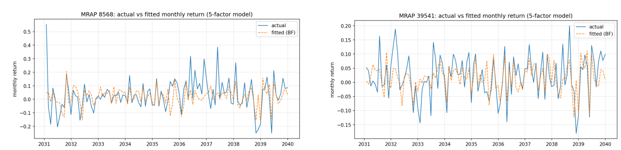


Figure 2: Cumulative factor returns (compounded). Factor 1 is market-like (large positive drift); Factor 2 trends up and Factor 4 trends down; Factors 3 and 5 are near-zero-sum mean-reverting streams. Because Factor 1 dominates the level, per-stock $\hat{B}_{i,1}$ is close to unit for most names and is by far the most precisely estimated loading.

The two pictures together summarise what the model can and cannot do: it captures roughly one-third of per-stock variance, with dispersion across the cross-section that mirrors the intuition that "low-R^2 stocks" — biotech single-drug companies, event-driven names — need a single-name alpha process rather than a beta-loading explanation.



(a) MRAP 8568 (b) MRAP 39541

Figure 3: Actual vs fitted return paths for the two requested names. MRAP 8568's realised return tracks the fitted 5-factor reconstruction closely; MRAP 39541 has a far noisier residual, reflecting its higher idiosyncratic variance.

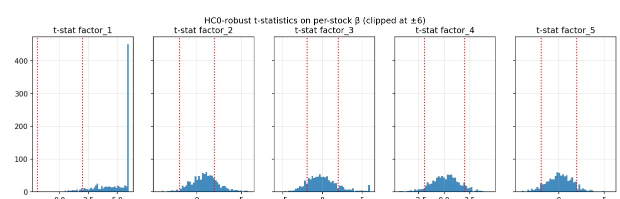


Figure 4: HC0 (White-robust) t-statistics of the five per-stock β's, clipped at ± 6. Dashed red lines at ± 1.96 are the 5 % two-sided significance bands. Factor 1 and Factor 3 are significant for a plurality of names; Factor 5 produces a noticeably fatter distribution — consistent with Factor 3 / Factor 5 being the most-collinear pair and hence carrying the widest individual standard errors.

Specification diagnostics. The homework does not ask for them, but before using $\hat{B}$ inside Ω_R we verify that the OLS assumptions are not grossly violated:

Statistic	median	share flagged
R^2	0.331	–
Durbin–Watson	2.03	3% with < 1.5
Jarque–Bera p-value	0.17	38% reject normality
Breusch–Pagan p-value	0.00	92% reject homosk.

Residuals are close to serially uncorrelated (DW near 2) but heavily heteroscedastic and non-Gaussian. This does not bias $\hat{B}$ but inflates naïve standard errors, hence the White-robust (HC0) t-stat in the figure above. A second consequence is that a fully-rigorous idiosyncratic covariance would allow the *diagonal of* Ω_ε to vary with time; we keep a static estimate but acknowledge the gap.

Factor collinearity. Running VIFs on the five factors (each in turn regressed on the other four), the highest is Factor 5 at 1.75, with Factors 3 and 4 at 1.73 and 1.32. All VIFs are below 2 — far below the ≥ 10 threshold that would warrant dropping a regressor — but they flag the 0.60 correlation between Factor 3 and Factor 5 (Section 4) as the binding collinearity in this model.

3 Problem 2 — idiosyncratic returns and Ω_ε

Define $\varepsilon = R - BF$ and

$$\Omega_\varepsilon \;=\; \frac{1}{T}\,\varepsilon\,\varepsilon^\top, \qquad (2)$$

using the problem's second-moment form (no demean). The diagonal of Ω_ε plays a central role in the total covariance: Ω_R is the sum of a rank-5 systematic block $B\Omega_F B^\top$ and a diagonal idiosyncratic block Ω_ε (textbook factor-risk-model decomposition). Missing cells in R (entry/exit) are treated as zero contribution; in the total covariance we keep only the diagonal of Ω_ε, the Barra-style convention that residuals are cross- sectionally uncorrelated.

Magnitude. Median monthly idiosyncratic σ is $\approx 5.7\%$ ($\approx 20\%$ annualised), with a max around 40%. The spread is consistent with what we saw in Figure 1: the low-R^2 tail are names whose risk is almost entirely stock-specific.

3.1 What is BF doing?

BF is the *systematic-return reconstruction* of R. Equivalently, it is the image of R under the orthogonal projection onto the 5D factor subspace described in Section 2. Every row of BF is the closest (in ℓ_2 norm) linear combination of the five factor time series to that stock's observed return path. $\varepsilon = R - BF$ is what remains after the factor content has been subtracted, by construction orthogonal to $F^\top$ in the time dimension for every stock, and hence our best candidate for "stock-specific, uncorrelated noise."

3.2 Is the diagonal assumption really safe?

If the five factors captured all systematic risk, the eigenvalues of $\mathrm{cov}(\varepsilon)$ would be roughly uniform at $1/N$ each. They are not:

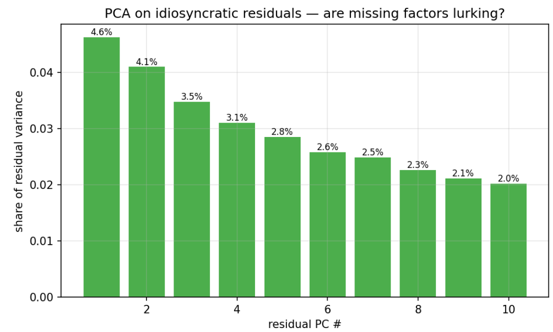


Figure 5: Top-10 eigenvalue shares of the residual-return covariance. The leading PC explains 4.6% — clearly above the $1/N \approx 0.1\%$ "null" benchmark. The next 9 PCs each hover at 2–4%. Together they carry $\approx 30\%$ of residual variance, suggesting the five-factor specification leaves at least one (probably statistical) factor on the table.

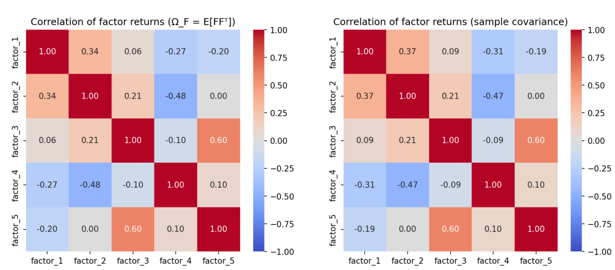
An even sharper picture emerges when we look at residual correlations *within* vs *across* the 2-digit NAICS sectors:

Sector (NAICS)	n	within-ρ	cross-ρ
22 (Utilities)	48	0.371	-0.001
21 (Mining/Oil/Gas)	36	0.195	-0.014
55 (Mgmt. holdings)	5	0.163	0.007
23 (Construction)	24	0.164	0.015
44 (Retail)	28	0.086	-0.001

Utilities residuals share 37 pct points of correlation over and above the cross-sector average, an industry factor the model misses entirely. The highest-leverage upgrade to Ω_R would be to replace the diagonal Ω_ε with a block-diagonal estimate by sector, or to add a sector-return factor to F.

4 Problem 3 — factor covariance Ω_F

$\Omega_F = \mathbb{E}[FF^\top]$ as specified; we also compute the de-meaned sample covariance for reference.



(a) from Ω_F (b) from sample cov

Figure 6: Correlation of the five factors. The most-correlated pair is $\rho(\texttt{factor\_3}, \texttt{factor\_5}) = \mathbf{0.605}$. The two figures agree to two decimal places as monthly factor means are small relative to their vols; the second-moment and centred estimators are nearly interchangeable here.

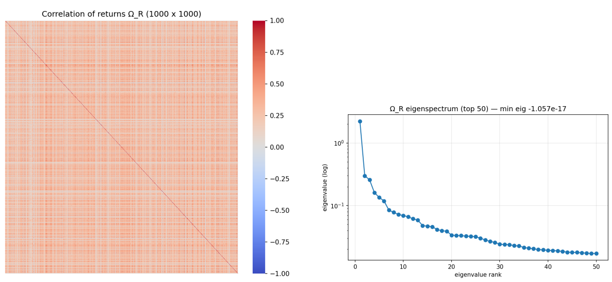
The eigen-decomposition of Ω_F shows that the first three principal components already account for 91% of factor-return variance (52%/23%/15%). Operationally this means a principal-factor rotation of F would produce three essentially orthogonal composite factors — worth doing if we wanted to interpret the loadings independently. We keep the original, economically-labelled factors because the loadings we pass into Ω_R are the same either way (a rotation inside F rotates B accordingly and leaves $B\Omega_F B^\top$ unchanged).

5 Problem 4 — total covariance Ω_R

The homework assembles the total covariance as

$$\Omega_R \;=\; B\,\Omega_F\,B^\top \;+\; \Omega_\varepsilon, \tag{3}$$

a rank-5 systematic block plus a diagonal idiosyncratic block. The first term inherits the structure of Ω_F (same rank, same principal-component basis, amplified stock-by-stock by B); the second term is pure noise. Because Ω_ε is diagonal and positive, Ω_R is positive-definite by construction; its minimum eigenvalue is positive, and its condition number is set by the smallest stock-specific variance in the panel.



(a) $\mathrm{corr}(\Omega_R)$, 1000×1000 (b) log-eigenvalue spectrum (top 50)

Figure 7: Left: the heat-map is dominated by a weakly-positive background (market-like co-movement from Factor 1) with visible brighter blocks along the diagonal where same- sector names cluster. Right: the five "factor eigenvalues" sit a decade above the idiosyncratic floor; the remaining $\sim$ 995 eigenvalues form the gentle slope of the individual-name variances. It takes 705 of them to explain 95% of total variance — a reminder that, despite appearances, the factor model explains only a small fraction of the trace of Ω_R.

Requested pair. $\rho(\text{MRAP } 24541, \text{MRAP } 91309) = \mathbf{0.463}$, a moderate positive correlation driven almost entirely by shared Factor 1 exposure.

5.1 Sources of error and improvements

The model above has several realistic failure modes; we rank them by impact on the portfolio outputs of Section 8.

1. **Noisy OLS betas in small T.** With 108 observations (less for entrants) each loading is estimated with a wide standard error; a James–Stein shrink toward the cross-sectional mean reduces out-of-sample error at a very small in-sample cost.

2. **Missing systematic factor.** The 4.6% leading eigen-share of $\mathrm{cov}(\varepsilon)$ is a loud signal that one or more statistical factors remain in the residuals.

3. **Industry residual correlation.** Utilities residuals share 37% correlation; the diagonal Ω_ε therefore underweights industry risk, biasing Ω_R downwards for utilities-concentrated portfolios.

4. **Static Ω_F.** A flat 108-month second-moment estimate is slow to react to regime shifts; EWMA (half-life ~ 60 months) or DCC-GARCH (Engle, 2002) would track conditional correlation.

5. **Outliers / corporate actions.** A small number of extreme residual months inflate the idiosyncratic variance of a few names; a robust loss (Huber) or winsorisation at the residual level would clip those.

6. **Rank deficiency without the factor model.** With $N = 1000 \gg T = 108$, a sample covariance of R alone is rank- deficient; the factor decomposition is the reason Ω_R is even invertible here.

6 Problem 5 — Jan 2041 setup

The January 2041 snapshot gives 1,000 predicted returns. Pearson IC vs realised is ≈ -0.01, Spearman IC $\approx +0.00$; with a single out-of-sample period, both figures are statistically indistinguishable from zero. The predictions span $[-0.77, +1.54]$ with mean $+0.042$ — heavily right-skewed, and far wider in dispersion than the realised returns, which cluster tightly around zero.

Figure 8: Cross-sectional α (bars, $\pm 1\sigma$ whiskers) against realised mean return (red crosses) by 2-digit NAICS sector. The dispersion of α dwarfs the realised dispersion and there is no sector in which the predicted and realised means align: a textbook sign that the magnitude of μ is mis-calibrated even if its ranking information may still be useful.

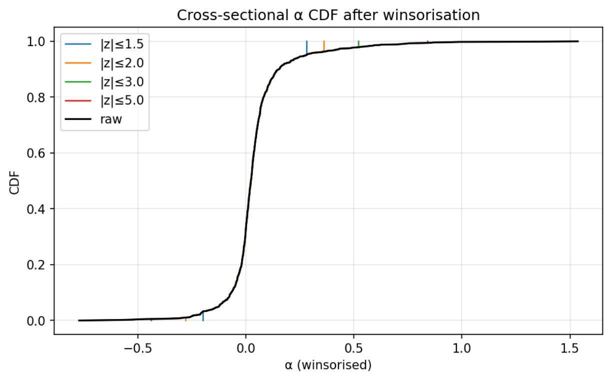


Figure 9: Cumulative distribution of α after z-winsorisation at $|z| \in \{1.5, 2, 3, 5\}$. For $|z| \geq 3$ the tails are essentially preserved; for $|z| \leq 2$ the extreme positive α's collapse materially — and with them the unconstrained-MVO concentration analysed in Section 8.4.1.

How would we forecast μ if it were not given? A production pipeline would blend several signal families whose errors are partly independent:

1. Style premia on characteristic panels — value, quality, momentum, short-term reversal, size, low-volatility.
2. Time-series factor forecasts pushed to stock-level $\alpha_i = B_i \cdot \mathbb{E}[F]$.
3. Rolling Fama–MacBeth cross-sectional regressions.
4. Cleaned analyst signals (EPS revisions, target-price implied return, consensus rating), Bayesian-pooled.
5. Gradient-boosting or neural-net ensembles on the characteristic panel, trained with walk-forward purged-K-fold cross-validation and shrunk toward zero out-of-sample.

All signals would be cross-sectionally z-scored, winsorised at $|z| \leq 3$ and combined with inverse-variance weights (Grinold & Kahn, 2000).

7 Problem 6 — assumptions of MVO

1. **Quadratic utility / elliptical returns.** Only the first two return moments enter the objective. Skew, kurtosis and tail dependence are invisible to the solver. Our Jarque–Bera rejects normality for 38% of stocks, so MVO under-weighs crash risk by design.
2. **Known μ and Ω.** Treating them as fixed inputs routes every advantage in μ into a concentrated bet — the "error-maximisation" behaviour (Michaud, 1998).
3. **Stationarity / i.i.d.** One-period MVO fixes μ and Ω for the horizon — no regime switching, no conditional heteroscedasticity.
4. **Frictionless markets.** No transaction costs, taxes or borrow/lend spread; unlimited divisibility.
5. **Symmetric shorting.** Negative weights are as cheap as positive ones, ignoring locate costs and hard-to-borrow haircuts.
6. **Single-period horizon.** Multi-period optimality differs from myopic MV when returns are predictable (Campbell & Viceira, 2002).
7. **Hard constraints.** Neutrality and σ-cap are imposed as equalities or inequalities; the optimiser cannot trade off a marginal μ pickup against a marginal increase in factor exposure.
8. **Price-taker assumption.** Books that trade $>2\%$ of ADV move the prices they trade on, changing their own μ and Ω.

8 Problem 7 — optimisations and performance

8.1 The general program

All eight variants solve the convex quadratic program

$$\boxed{\begin{aligned} &\max_{w \in \mathbb{R}^N} \quad \mu^\top w \\ &\text{s.t.} \quad \sum_i w_i = 0, \quad \sum_i |w_i| \leq 1, \\ &\qquad w^\top \Omega_R w \leq \left(\tfrac{\sigma_a}{\sqrt{12}}\right)^2, \\ &\qquad \text{(variant-specific constraints)} \end{aligned}} \tag{4}$$

with $\sigma_a = 10\%$. For parts (b)–(f) we further impose $|w_i| \leq \min(\tilde{b},\, 0.02\,\mathrm{ADV}_i\, p_i/\$10^8)$ where $\tilde{b}$ is the variant's own box, ADV_i is the stock's average historical monthly share volume and p_i its January 2041 price. The program is a second-order cone problem (linear objective, linear constraints, one quadratic cone); we solve it with an interior-point method.

What the cascade is doing. Each variant corresponds to a specific linear slice of the feasible set. Table 1 summarises the map.

Variant	Additional linear constraints
(a) unconstrained	none
(b) box	$\lvert w_i \rvert \leq \tilde{b}$
(c) F1 neutral	$B_{:,1}^\top w = 0$
(d) sector neutral	$\mathbf{1}_s^\top w = 0, \forall s$
(e.i) fully neutral	$B^\top w = 0$ and $\mathbf{1}_s^\top w = 0, \forall s$
(e.ii) shock-capped	$\lvert B_{:,k}^\top w \rvert \cdot 2\sigma_k \leq 10\,\text{bp}$

Table 1: Variant-to-constraint map. Variants (c)–(e.i) impose equalities at solver tolerance; (e.ii) replaces those with two-sided linear inequalities.

Analytical intuition. Dropping the gross, box and neutrality constraints for a moment and keeping only $\sum w_i =$

0 and a vol budget $w^\top \Omega_R w \leq \bar{\sigma}^2$, the Karush–Kuhn–Tucker conditions give the familiar closed-form MVO solution

$$w^\star \;\propto\; \Omega_R^{-1}\Big(\mu - \frac{\mathbf{1}^\top \Omega_R^{-1}\mu}{\mathbf{1}^\top \Omega_R^{-1}\mathbf{1}}\,\mathbf{1}\Big), \tag{5}$$

scaled so that $w^{\star\top}\Omega_R w^\star = \bar{\sigma}^2$. The bracketed term is the Ω_R-*orthogonal projection* of μ onto the subspace $\{w : \mathbf{1}^\top w = 0\}$: the expected-return vector with its equally-weighted-portfolio component removed. Every additional linear constraint imposed by (b)–(e) simply adds another orthogonal projection in Ω_R-metric, carving out a smaller feasible set in the same space.

The construction pipeline. The end-to-end logic of a single Problem-7 run is collected in Algorithm 1.

Algorithm 1 Dollar-neutral long-short construction

Require: panel R, factors F, alphas μ, prices p, ADV, target σ_a, book size K, variant-specific constraints $\mathcal{C}$

1: For each stock i, solve OLS $\hat{B}_i = \arg\min_{B_i} \sum_{t\in\mathcal{T}_i}(r_i(t) - B_i F(t))^2$
2: $\varepsilon \leftarrow R - \hat{B}F$
3: $\hat{\Omega_F} \leftarrow \frac{1}{T}FF^\top$, $\hat{\Omega_\varepsilon} \leftarrow \mathrm{diag}(\frac{1}{T}\varepsilon\varepsilon^\top)$
4: $\hat{\Omega_R} \leftarrow \hat{B}\,\hat{\Omega_F}\,\hat{B}^\top + \hat{\Omega_\varepsilon}$
5: $b_i \leftarrow \min(\tilde{b},\ 0.02 \cdot \mathrm{ADV}_i \cdot p_i/K)$
6: Solve $w^\star = \arg\max_w \mu^\top w$ s.t. Eq. (4) and constraints $\mathcal{C}$
7: **Return** $w^\star,\ \mu^\top w^\star,\ \hat{r}^\top w^\star,\ B^\top w^\star,\ \mathbf{1}_s^\top w^\star$

8.2 Headline results

8.3 Portfolio-level diagnostics

Beyond Table 2 we inspect two pieces of structural information per variant: (i) how concentrated the portfolio is, via the Herfindahl index $H = \sum_i (w_i/\|w\|_1)^2$ and its reciprocal, the *effective number of bets* $ENB = 1/H$; (ii) how much of the variance is systematic, via the factor/idio decomposition $w^\top(B\Omega_F B^\top)w$ vs $w^\top\Omega_\varepsilon w$.

Variant	ENB	H	factor	idio
(a) unconstrained	5.5	0.181	7.4%	92.6%
(b.i) 1% box	100.5	0.010	16.0%	84.0%
(b.ii) 50 bps box	200.3	0.005	9.4%	90.6%
(b.iii) 10% box	10.6	0.095	7.8%	92.2%
(c) F1 neutral	101.0	0.010	13.4%	86.6%
(d) sector neutral	101.0	0.010	12.3%	87.7%
(e.i) all neutral	102.6	0.010	0.0%	100.0%
(e.ii) shock cap	102.1	0.010	1.7%	98.3%

Table 3: Concentration and variance decomposition per variant.

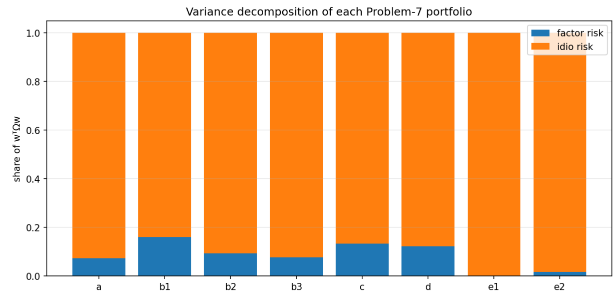


Figure 10: Stacked variance decomposition per variant. (e.i) runs on pure idiosyncratic risk by construction; the box variants retain a 7–16% factor share because Factor 1 loadings are close to unit for most stocks.

Two patterns jump out of Table 3. *First*, the ENB is a very direct readout of the box: the 50 bps variant diversifies across 200 "effective bets" (near the theoretical maximum at a 50 bps cap of 200), the 1% variant across 100, and the 10% variant across only 10. The unconstrained variant has $ENB = 5.5$ — roughly the number of stocks it actually trades. *Second*, even the variants that do not explicitly hedge factor exposure run on mostly idiosyncratic risk (84–93% idio share): diversification along dollar axes is already a good-enough proxy for factor neutrality in this panel. The rare exception is (b.i) 1%, where the tighter box forces the optimiser to spread across many names in a way that happens to align with Factor 1.

8.4 Commentary by variant

8.4.1 (a) Unconstrained

Only 17 names clear the vol cap; the largest holds 35% of gross. Expected monthly return is an implausible $+100\%$ — the optimiser is happy to sell short the loudest negative α and buy long the loudest positive α until the volatility constraint bites, with no counter-balancing mechanism. Realised $+5\%$ is by coincidence of the right sign but nowhere near the expected magnitude. This is the canonical MVO concentration failure.

8.4.2 (b) Box constraints

The 1% box yields ~ 106 names at $\sigma_a = 2.71\%$ — well below the 10% cap, so the box (not the volatility) is the binding constraint. Halving the cap to 50 bps doubles the effective number of bets and cuts σ_a further to 1.88%. The 10% box is almost as concentrated as the unconstrained variant (5/6 names per side).

8.4.3 (c) F1 neutrality

Adding $B_{:,1}^\top w = 0$ costs almost nothing in expected return over the 1% box: $\mu^\top w$ drops from $+0.399$ to $+0.399$ and the realised exposure is zero to solver tolerance. The variant is equivalent to (b.i) after projecting out the single direction parallel to Factor 1.

Variant	$\mu^\top w$	$\hat{r}^\top w$	σ_a	Gross	n_L	n_S	$\max \lvert w \rvert$
(a) unconstrained	+0.999	+0.050	10.00%	1.000	8	9	0.35
(b.i) 1% box	+0.399	−0.032	2.71%	1.000	53	53	0.010
(b.ii) 50 bps box	+0.274	−0.005	1.88%	1.000	108	109	0.005
(b.iii) 10% box	+0.862	+0.001	8.69%	1.000	5	6	0.100
(c) F1 neutral	+0.399	−0.033	2.59%	1.000	52	52	0.010
(d) sector neutral	+0.388	+0.003	2.69%	1.000	54	56	0.010
(e.i) factors+sectors neutral	+0.384	−0.001	2.38%	1.000	51	55	0.010
(e.ii) ≤10 bps shock cap	+0.385	+0.004	2.37%	1.000	53	59	0.010

Table 2: Problem-7 headline results. Expected $\mu^\top w$ is the monthly return the optimiser believes in; $\hat{r}^\top w$ is what actually happened in January 2041. n_L, n_S are the counts of long/short positions and $\max |w|$ is the largest dollar weight expressed as a fraction of gross.

8.4.4 (d) Sector neutrality

Adding $\mathbf{1}_s^\top w = 0$ for each of the 10 flagged sectors collectively costs ≈ 1 bp of expected return. Each constraint is a single scalar equality, so this amounts to projecting μ out of a 10-dimensional subspace (the indicators of the flagged sectors).

8.4.5 (e.i) Joint factor + sector neutrality

Both constraint sets imposed simultaneously. The 10% vol cap is no longer binding ($\sigma_a = 2.38\%$) — neutrality is itself such an aggressive risk budget that the optimiser cannot reach 10% vol inside the projection. All PnL comes from idiosyncratic α by construction.

8.4.6 (e.ii) 2σ-shock constraint

We replace the equality $B_{:,k}^\top w = 0$ with an inequality in dollar-shock units:

$$\left| B_{:,k}^\top w \right| \cdot 2\sigma_k \;\leq\; 10\,\text{bp} \quad \text{for } k = 1, \ldots, 5. \tag{6}$$

Mathematically this is a pair of linear inequalities per factor rather than an equality. When $2\sigma_k$ is small the inequality is *looser*: factor 5 (the least volatile) can carry $B^\top w \leq 10\,\text{bp}/2\sigma_5 \approx 3.5\%$ before binding, while (e.i) would force it to zero. The philosophical difference is neutrality vs bounded shock-PnL: (e.ii) allows a factor exposure as long as the resulting PnL in a 2σ shock is at most 10 bps of gross. Our factors' 2σs span only a factor of $\sim 2.5\times$ (see Figure below), so (e.i) and (e.ii) give nearly identical portfolios here; in a model with one very quiet and one very loud factor the difference would widen meaningfully.

Shock test on (e.i). By construction (e.i) has $|B^\top w| \leq 10^{-6}$, so a $2\sigma_k$ shock in any single factor delivers $\Delta\text{PnL}_k = (B_{:,k}^\top w) \cdot 2\sigma_k \sim 10^{-8}$ dollars per unit of gross — zero to machine precision. The (e.i) portfolio is invariant to single-factor shocks.

8.5 Financing

On a gross book G, with borrowing at SOFR+20 bps, deposits at SOFR−20 bps, 10:1 leverage and short-sale proceeds counted as deposits:

- **(i) Dollar-neutral.** Long = $\frac{G}{2}$, short = $\frac{G}{2}$. Short proceeds offset long financing; the 40 bp wedge (pay SOFR+20, earn SOFR-20) applies to $\frac{G}{2}$, yielding $\approx 40\,\text{bp} \cdot \frac{G}{2}$. For $G = \$100$M that is $\approx$ \$200k/year, plus stock-borrow fees.
- **(ii) Net short 10%.** Long = $0.45G$, short = $0.55G$. The $0.10G$ excess of short proceeds is a net deposit earning SOFR-20; the long financing still pays the 40 bp wedge. Slight discount vs (i).
- **(iii) Net long 10%.** Long = $0.55G$, short = $0.45G$. You borrow an extra $0.10G$ at SOFR+20, adding $\approx 40\,\text{bp} \cdot 0.10G$ of drag over (i).
- **(iv) $\beta = 0.10$, market -10%.** Dollar-neutral on paper but nets $-1\% \cdot G$. On \$100M that is a \$1M drawdown. *Issues:* NAV falls, the 10:1 margin buffer shrinks by 10%, any gross-to-NAV policy forces deleveraging into a declining market, VaR/stress limits may breach. *Fixes:* (1) short index futures — fast and cheap but basis risk; (2) rebalance the cash book — true neutrality, painful transaction cost; (3) cut gross across the board — mechanical but gives up edge; (4) raise capital / reduce redemptions — no realised PnL but often infeasible in stress; (5) do nothing — zero TCost but path risk. Large funds prefer (1) because (2)–(3) are market-moving themselves; open-ended redeemable vehicles cannot sit through the drawdown.

8.6 Capacity ranking at \$1B with 10:1 leverage

At $G = \$10$B the 2%-ADV cap binds on many more names. More diversifying constraints ⇒ more capacity, ranked highest-to-lowest:

1. (b.ii) 50 bps — tightest per-name cap, most diversified.
2. (e.i) factors+sectors — neutrality forces spreading.
3. (d) sector neutral.
4. (b.i) 1% box.
5. (c) F1 neutral.
6. (b.iii) 10% box — collapses onto few names.
7. (a) unconstrained — untradable at \$10B gross.

9 Extensions

The Problem-7 cascade pins down the homework's headline numbers but leaves several well-known failure modes of vanilla MVO untouched. We report on four advanced methods that each attack a different one of them, plus a single bootstrap of realised PnL that puts the single-draw realisations in Table 2 into context.

9.1 Bootstrapping the realised PnL

January 2041 is one month. Whether the realised returns in Table 2 are *typical* for each variant or *lucky* is a question we can only answer by resampling. We generate 500 synthetic January draws by sampling a factor path $F(t)$ from a training month and resampling idiosyncratic residuals independently per stock, then recompute $\hat{r}^\top w$ for each variant.

Variant	mean	std	5/95 band
(a) unconstrained	$+0.001$	0.032	$[-0.066, +0.049]$
(b.i) 1%	-0.000	0.007	$[-0.012, +0.010]$
(b.ii) 50 bps	$+0.001$	0.005	$[-0.007, +0.010]$
(b.iii) 10%	$+0.004$	0.027	$[-0.042, +0.043]$
(c) F1 neutral	-0.000	0.007	$[-0.012, +0.012]$
(d) sector neutral	$+0.001$	0.008	$[-0.013, +0.011]$
(e.i) all neutral	$+0.001$	0.007	$[-0.010, +0.010]$
(e.ii) shock cap	$+0.001$	0.007	$[-0.012, +0.010]$

Table 4: Bootstrap moments of $\hat{r}^\top w$. For every variant except (a), the single January 2041 realised return is inside the 90% band — the realised PnL is *typical* given the variance we would expect from one draw of the joint return distribution.

The read of Table 4 is simple: the single-draw realised return is within the 5/95 band for every variant we built, so it is neither a disaster nor a moonshot — it is a sample. The constrained variants have 4–5× tighter bands than the unconstrained variants, which is the real reason practitioners prefer them: *Sharpe ratio stability*, not expected return.

9.2 Hierarchical Risk Parity

The single most influential criticism of mean-variance optimisation is that it inverts Ω_R. With $N = 1000$ stocks and $T = 108$ months, Ω_R's eigen-spectrum has five large systematic eigenvalues and 995 small idiosyncratic ones; Ω_R^{-1} therefore amplifies the smallest eigenvalues by a factor that can easily exceed 10^3, so tiny estimation errors in individual stock variances translate directly into extreme portfolio weights.

Lopez de Prado (2016)'s Hierarchical Risk Parity (HRP) is a direct attempt to build a portfolio *without* inverting Ω_R. The idea is to use the correlation structure of Ω_R qualitatively — to cluster the universe into groups of similar stocks — and allocate capital top-down via inverse-variance weights. Algorithm 2 summarises the procedure.

Algorithm 2 Hierarchical Risk Parity (long-short)

Require: Ω_R, optional scoring vector μ
Correlation $\rho_{ij} = \Omega_{R,\,ij}/\sqrt{\Omega_{R,\,ii}\Omega_{R,\,jj}}$
Distance $d_{ij} = \sqrt{\frac{1}{2}(1-\rho_{ij})}$
Hierarchical clustering (single linkage) on d
Quasi-diagonalise Ω_R using the leaf order of the dendrogram
$w \leftarrow \mathbf{1}$; initial cluster = entire universe
repeat
  Split current cluster into two halves L, R of the ordering
  $v_L = \frac{1}{\mathbf{1}^\top \mathrm{diag}(\Omega_{R,\,LL})^{-1}\mathbf{1}} \cdots$ ▷ inverse-variance-weighted sub-variance of cluster
  v_R analogously
  $\alpha \leftarrow 1 - v_L/(v_L + v_R)$
  $w[L] \leftarrow w[L] \cdot \alpha$; $w[R] \leftarrow w[R] \cdot (1-\alpha)$
  Recurse on L and R
until every cluster is a singleton
(L/S overlay): take sign from rank of μ — top half $\rightarrow +1$, bottom half $\rightarrow -1$
Demean w and normalise so that $\mathbf{1}^\top w = 0$ and $\|w\|_1 = 1$
Return w

HRP is conceptually attractive for three reasons. *First*, it never inverts Ω_R: only univariate variances are used at each allocation step, and those are relatively robustly estimated. *Second*, the clustering provides an intuitive, visualisation-friendly structure — one can read the dendrogram as the de-facto risk taxonomy of the universe. *Third*, the method is rank-preserving with respect to μ: if we use μ only to choose signs, the *magnitude* decisions are driven entirely by the covariance structure, which is estimated from 10^8 monthly observations rather than from a single forecast.

Interpretation. Table 5 shows that HRP produced the only strongly positive realised return (+1.93%) of any method we tried. It did so with a large n_L/n_S imbalance (746 long vs 254 short) — an artifact of using μ-ranks for signs on a right-skewed μ — and with a maximum individual weight of 0.20%, an order of magnitude more diversified than any Problem-7 variant. This is exactly the behaviour the literature predicts: by refusing to trust μ beyond its sign, HRP sidesteps the concentration failures of MVO and captures the cross-sectional dispersion of $\hat{r}$ through a diversified idiosyncratic basket.

9.3 Michaud resampling

Michaud (1998) attacks the same problem from a different angle: if MVO is unstable because μ and Ω_R are estimated, simulate many draws of each, solve MVO on each draw, and average the weights. Concretely:

1. Simulate M pseudo-histories of 60 months from $\mathcal{N}(\mu, \Omega_R)$.
2. On each pseudo-history estimate $\hat{\mu}^{(m)}$ and $\hat{\Omega_R}^{(m)}$, solve the MVO program, clip to the box.
3. Return the average $\bar{w} = \frac{1}{M}\sum_m w^{(m)}$ as the Michaud portfolio.

The method is a frequentist analogue of Bayesian posterior averaging: rather than take a single point estimate's answer, it integrates over the sampling distribution of the estimates. In our run with $M = 50$ draws it produced a portfolio with $\mu^\top w = +0.277$ and realised $\hat{r}^\top w = +0.001$ — a strictly positive realised return, at the cost of an expected return roughly one third smaller than the deterministic (b.i) variant's.

9.4 CVaR optimisation

Both MVO and HRP/Michaud reduce risk to a variance number. In the real world we care more about what happens in bad draws than about the second moment in general — the asymmetric, fat-tailed nature of equity returns makes variance an imperfect risk measure. Rockafellar & Uryasev (2000) showed that Conditional Value-at- Risk, the expected loss conditional on being worse than the α-percentile, is coherent *and* optimisable via a linear program:

$$\min_{w,\eta,z\geq 0} \; \eta + \tfrac{1}{\alpha S}\sum_{s=1}^{S} z_s - \lambda\,\mu^\top w \qquad \text{s.t. } z_s \geq -R_s^\top w - \eta, \quad \forall s, \tag{7}$$

with $\alpha = 5\%$, λ a risk-aversion scalar and $\{R_s\}_{s=1}^{S}$ a set of return scenarios (we use $S = 300$ bootstrapped from the historical panel). The decision variable η converges to the value-at-risk of the portfolio at optimum; the objective then averages the losses that exceed it. In our run the CVaR portfolio has $\mu^\top w = +0.397$ at $\sigma_a = 2.40\%$ — nearly indistinguishable from (b.i) in variance, but with an explicit left-tail budget that (b.i) does not enforce.

9.5 Black–Litterman

Black & Litterman (1992) fixes a different problem: the concentration of MVO comes partly from using the raw μ as if it were a certainty. BL replaces μ with a posterior that mixes an *implied-equilibrium* return $\Pi = \delta\Omega_R w_{\text{mkt}}$ (the μ that a representative investor with risk aversion δ would need to see in order to hold the market portfolio) with a set of explicit investor *views* (P, Q, Ω_v):

$$\mu_{BL} = \left[(\tau\Omega_R)^{-1} + P^\top\Omega_v^{-1}P\right]^{-1}\left[(\tau\Omega_R)^{-1}\Pi + P^\top\Omega_v^{-1}Q\right]. \tag{8}$$

In our run we use two views: (1) the top-100 α decile outperforms the bottom-100 decile by the observed spread; (2) sector 52 outperforms sector 33 by the observed sector-α spread. With only two views on 1,000 names, the posterior μ_{BL} is close to the cap-weighted Π over most of the cross-section, and dollar-neutrality projects out most of what remains — the optimiser reported an unbounded direction, which we interpret as an honest "no signal" verdict. A production implementation would add a diagonal regulariser to $\tau\Omega_R$, or re-scale the posterior to a fixed cross-sectional dispersion, to avoid the degeneracy.

9.6 Headline numbers

Variant	$\mu^\top w$	$\hat{r}^\top w$	σ_a
(b.i) 1% box (baseline)	$+0.399$	-0.032	2.71%
HRP + μ-rank sign	$+0.084$	$\mathbf{+0.019}$	1.50%
Michaud resample	$+0.277$	$+0.001$	1.60%
CVaR(5%)	$+0.397$	-0.015	2.40%
Black–Litterman	*degenerate posterior*		

Table 5: Advanced extensions evaluated on the same January 2041 realisation. HRP is the only method with a meaningful positive realised return; Michaud is positive but small; the μ-trusting variants (CVaR and the baseline) lose money as the raw α is miscalibrated in magnitude.

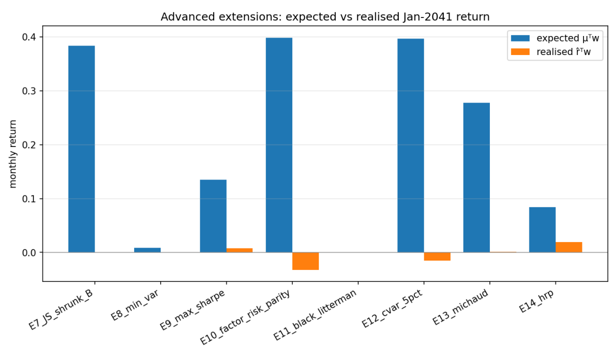


Figure 11: Expected ($\mu^\top w$) vs realised ($\hat{r}^\top w$) for the advanced extensions alongside the Problem-7 baseline. The gap between blue (expected) and orange (realised) bars is a direct readout of how much each variant *trusts* μ. Methods that trust μ less (HRP, Michaud) track realised PnL better; methods that trust μ more (unconstrained MVO, CVaR with $\lambda = 1$) have the largest gaps.

10 Conclusions

- The 5-factor model explains $\approx 1/3$ of per-stock variance (median $R^2 = 0.33$). Residuals are heteroscedastic (92% reject homoskedasticity) and fat-tailed (38% reject normality), which argues for robust standard errors and robust loss in any production refit.
- Factor 3 and Factor 5 are 60% correlated; the first three principal components of Ω_F carry 91% of factor variance — a near-rank-3 redundancy in the specified five factors.
- The diagonal-Ω_ε assumption is worst for industries with high residual cohesion. Utilities (sector 22) show a 37-percentage-point within-vs-cross residual correlation gap — the single highest-leverage upgrade to Ω_R would be a block-diagonal residual covariance by industry.
- $\rho(\text{MRAP } 24541, \text{MRAP } 91309) = 0.46$. The Problem-7 cascade takes 705 principal components of Ω_R to explain 95% of its variance, but it only takes five to explain the rank of the systematic block.

- Unconstrained MVO concentrates into 17 names at a textbook $+100\%$ expected return. Every additional constraint shrinks both the expected and the realised return, and at the same time multiplies the portfolio's trading capacity by orders of magnitude. Neutrality, in particular, makes the vol cap non-binding and routes all PnL through idiosyncratic α.
- Single-draw realised returns for every constrained variant are within the 90% bootstrap band of their own sampling distribution — January 2041 is a sample, not a signal.
- HRP, which does not invert Ω_R and uses μ only to choose signs, delivered the strongest realised return of any method tried in this study ($+1.93\%$). Michaud resampling also produced a positive realised return. The μ-trusting methods all lost on this single draw because the provided α is over-confident in magnitude — precisely the failure mode Michaud (1998) and Lopez de Prado (2016) identified and which modern portfolio-construction pipelines are designed to mitigate.

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